EXACTNESS OF THE SHEAFIFIED INVERSE IMAGE ALONG A COCONTINUOUS FUNCTOR

THE STACKS PROJECT AUTHORS

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The direct image along a cocontinuous functor is a sheaf. Same chapter, Lemma lemma-pu-sheaf. Supplies the functor on sheaves to which the adjunction is compared.
- The adjunction between the two functors on presheaves. Same chapter, Lemma lemma-adjoints-pu. Supplies the middle step of the chain of bijections.
- Left exactness of sheafification and of the inclusion. Same chapter, Lemma lemma-sheafification-exact and Lemma lemma-limit-sheaf. Show that the composite is left exact.
- Exactness of left adjoints. Categories, Lemma categories-lemma-exact-adjoint. Gives right exactness of the functor.

Source reference. Original lemma 00XK: sources/stacks-sites/pinned-original-source.tex, source lines 4568–4568. Navigation pointer invoked at proof line 4568; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 00W2: sources/stacks-sites/pinned-original-source.tex, source lines 4593–4593. Navigation pointer invoked at proof line 4593; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ and $\mathcal{D}$ be sites. Let $u : \mathcal{C} \to \mathcal{D}$ be cocontinuous. The functor $Sh(\mathcal{D}) \to Sh(\mathcal{C})$, $\mathcal{G} \mapsto (u^p\mathcal{G})^\#$ is a left adjoint to the functor ${}_su$ introduced in Lemma 00XK above. Moreover, it is exact.*

Proof. Let us prove the adjointness property as follows

$$\begin{aligned} \mathrm{Mor}_{Sh(\mathcal{C})}((u^p\mathcal{G})^\#, \mathcal{F}) &= \mathrm{Mor}_{PSh(\mathcal{C})}(u^p\mathcal{G}, \mathcal{F}) \\ &= \mathrm{Mor}_{PSh(\mathcal{D})}(\mathcal{G}, {}_pu\mathcal{F}) \\ &= \mathrm{Mor}_{Sh(\mathcal{D})}(\mathcal{G}, {}_su\mathcal{F}). \end{aligned}$$

Thus it is a left adjoint and hence right exact, see Categories, Lemma 0039. We have seen that sheafification is left exact, see Lemma 00WJ. Moreover, the inclusion $i : Sh(\mathcal{D}) \to PSh(\mathcal{D})$ is left exact by Lemma 00W2. Finally, the functor u^p is left exact because it is a right adjoint (namely to u_p). Thus the functor is the composition ${}^\# \circ u^p \circ i$ of left exact functors, hence left exact. □